Subject to change: Agreement and case

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Abstract

In contemporary colloquial Hebrew, unaccusative verbs may fail to exhibit ϕ-agreement with their subject when the latter appears postverbally, but not when it appears preverbally. To resolve disagreements regarding the role that intervention may play in licensing this lack of agreement, we conducted two acceptability judgment experiments, whose design and results we report. The findings indicate that intervention is not the factor licensing lack of agreement, contra Preminger's (2009, 2014) claim, which means the phenomenon does not corroborate his approach to grammaticality. Further, we provide evidence that this lack of ϕ-agreement with a postverbal internal subject results from the reanalysis of the subject as caseless. This is liable to occur in languages that do not morphologically mark nominative. Hebrew existential and possessive constructions had undergone a parallel process in the beginning of the past century.

Keywords: unaccusative, agreement, case, postverbal subject, existential and possessive constructions, Hebrew

1 Introduction

Agreement asymmetries between Subject-Verb (SV) word order and Verb-Subject (VS) order have often been discussed in the literature, in particular, with regard to Standard Arabic. Various approaches have been proposed to explain why in Standard Arabic SV order the verb shows full subject agreement, while in VS it fails to exhibit number agreement. The present paper explores a different kind of agreement asymmetry observed in contemporary colloquial Hebrew between SV and VS orders. In Hebrew, the default word-order is SV(Object), yet VS order is also permitted in several contexts. One such context involves verbs whose subject is an internal argument, namely, passives and unaccusatives (Reinhart and Siloni, 2005; Shlonsky, 1997). In contemporary colloquial Hebrew, such VS examples may lack

ϕ-agreement between the verb and its internal argument: the verb can appear with default ϕ-features (i.e. 3M.SG), even when the subject is feminine, plural, or both, as illustrated by the attested examples (1a-1c). This lack of ϕ-agreement is impossible in the corresponding SV counterparts (2a-2c). [1] Examples labeled (Conv.) are taken from conversational spoken Hebrew, a link is provided for online examples, and unmarked examples are constructed.

(1) a. *nigmar* *ha-tut-im.*
finished.3M.SG the-strawberry-M.PL
'There are no more strawberries.' (`https://tinyurl.com/2mdjatax`)

b. *šama hitxil* *ha-hadbaka* *ha-gdol-a.*
there began.3M.SG the-contagion.F.SG the-large-F.SG
'There the large-scale contagion began.' (Conv.)

c. *lo* *niš'ar* *mamtak-im* *ba-bayit.*
NEG remained.3M.SG candy-M.PL in.the-house
'There are no more candies in the house.' (`https://tinyurl.com/33xds6jv`)

(2) a. **ha-tut-im* *nigmar.*
the-strawberry-M.PL finished.3M.SG

b. **šama ha-hadbaka* *ha-gdol-a* *hitxil.*
there the-contagion.F.SG the-large-F.SG began.3M.SG

c. **mamtak-im lo* *niš'ar* *ba-bayit.*
candy-M.PL NEG remained.3M.SG in.the-house

While in Standard Arabic, partial agreement is obligatory in VS order, in colloquial Hebrew the phenomenon is optional and it involves no agreement at all; default ϕ-features surface. We briefly review the different types of accounts proposed for Standard Arabic, showing that they cannot be adapted to account for the Hebrew asymmetry.

Lack of ϕ-agreement in VS order in Hebrew is a relatively recent phenomenon. It has received little attention in the generative literature. In light of this and given that the phenomenon appears in low-register varieties of Hebrew, the exact VS configurations in which it is licensed are not yet agreed upon. The sole generative study that accounts for this lack of ϕ-agreement is Preminger (2009), who associates the phenomenon with intervention between the verb and the subject, a view that is inconsistent with attested examples, as will be shown below. To place our analysis on a solid empirical footing, we conducted two experiments, the results of which undermine the role of intervention.

We provide evidence that the lack of agreement with the internal argument subject in VS order in colloquial Hebrew results from the reanalysis of the postverbal subject as caseless

[1] (2c) should be read without topicalization intonation; with topicalization of *mamtakim*, the sentence is grammatical.

by certain speakers. This loss leads to a loss of agreement on the verb. Thus, two competing analyses are available to speakers for this VS configuration: (i) a case-bearing postverbal argument (subject) triggering ϕ-agreement; (ii) a caseless postverbal argument failing to trigger ϕ-agreement. We predict that the loss of case on the subject in the internal argument position followed by the loss of subject-verb agreement is likely to occur in languages that do not morphologically mark nominative, such as Hebrew.

The paper is structured as follows. Section 2 discusses previous approaches to the agreement asymmetries in Standard Arabic and colloquial Hebrew. Sections 3 and 4 present two experiments we conducted to examine the acceptability of lack of agreement in different types of configurations, and Section 5 develops our analysis. Section 6 concludes. We elaborate on the results of Experiment 2 in the appendix.

2 Previous approaches

2.1 Standard Arabic

In Standard Arabic, the verb shows full agreement with a preverbal subject (3a-3b), but fails to exhibit number agreement with its postverbal counterpart (4a-4b).

(3) Standard Arabic

a. *ṭ-ṭaalibaat-u ʔakal-na.*
the-student.F.PL-NOM ate-3F.PL
'The students ate.'

b. **ṭ-ṭaalibaat-u ʔakal-at.*
the-student.F.PL-NOM ate-3F.SG

(4) a. *ʔakal-at ṭ-ṭaalibaat-u.*
ate-3F.SG the-student.F.PL-NOM
'The students ate.'

b. **ʔakal-na ṭ-ṭaalibaat-u.*
ate-3F.PL the-student.F.PL-NOM (Ackema and Neeleman, 2012, 76:(1-2))

The postverbal subject can be either an external or an internal argument, it bears nominative case, and triggers partial agreement on the verb, failing to agree in number. Approaches differ as to whether the unrealized number feature was not there to begin with (e.g. Fassi Fehri (1988, 2013); Kinjo (2015)), or was generated but subsequently deleted or weakened at PF. Deletion has been argued to occur following verb-movement past the subject (Aoun et al., 1994). Weakening was proposed to result in a single realization of the number feature on the subject, either due to PF-merger (rebracketing under adjacency) of the postverbal subject and verb (Benmamoun, 2000; Benmamoun and Lorimor, 2006) or owing to a weakening rule (agreement weakening) applying within the prosodic phrase (Ackema and Neeleman, 2003, 2012).

The asymmetry in colloquial Hebrew is a different phenomenon and cannot be accounted for along similar lines. Unlike in Standard Arabic, the Hebrew asymmetry is not the result of an operation active in a restricted PF-domain, as the non-agreeing verb and its postverbal argument do not need to be adjacent or in the same prosodic phrase. In (5-7) a possessive dative intervenes between the two. Possessive datives are optional datives describing possession in the broad sense (ownership, authorship, responsibility etc.) over an internal argument (Borer and Grodzinsky, 1986; Plotnik et al., 2024).

(5) *nafal le-dani ha-maftexot.*
fell.3M.SG DAT-Dani the-key.M.PL
'Dani's keys fell.' (Preminger, 2009, 243(4b))

(6) *nafal la-yeled šeli ha-šen.*
fell.3M.SG DAT.the-kid.M my the-tooth.F.SG
'My son's tooth fell off.' (`https://tinyurl.com/mvnnz4m5`)

(7) *nigmar le-ima šeli ha-balonei gaz.*
finished.3M.SG DAT-mother my the-baloon.CS.M.PL gas
'My mom ran out of gas cylinders.' (`https://tinyurl.com/5xm7vynn`)

Further, unlike in Standard Arabic, in colloquial Hebrew ϕ-agreement is optional in VS order. When it is missing, its absence is total (not partial as in Standard Arabic), and there are no grounds to assume it was there to begin with; default ϕ-features surface. Finally, lack of agreement is possible only with the internal subject, not with the external one, although some configurations do permit postverbal external arguments in Hebrew. Specifically, external arguments can appear postverbally in the so-called 'triggered inversion' (Shlonsky, 1987), which is possible with any type of verb, as illustrated in (8a) with an unergative verb. This 'inversion' is licensed by an XP preceding the verb ('the trigger'), and appears mainly in high-register, written language. [2] Lack of ϕ-agreement with an external argument subject is impossible (8b).

[2]Shlonsky and Doron (1992) analyze the construction as a sort of 'verb second' phenomenon, with the subject in SpecTP. Borer (1995) locates the subject in its base position and the trigger in SpecTP.

As triggered inversion belongs to written, high-register varieties of Hebrew, it is not expected to exhibit lack of agreement, which is typical of low registers. In colloquial Hebrew, instances of VS order with an external-argument subject may occur when light material intervenes between the verb and its subject as in (i) ((Borer, 2005, 321); Meltzer-Asscher and Siloni (2012); Siloni (2002, 2012)). Lack of agreement in such constructions (ii) has not been documented (marked here with *).

(i) a. *omdim po ha-kisa'ot.*
stand.M.PL here the-chair.M.PL

(8) a. *ba-bkarim omd-im ha-robot-im be-šura.*
in.the-mornings stand-M.PL the-robot-M.PL in-line
'In the mornings the robots stand in line.'

b. **ba-bkarim omed ha-robot-im be-šura.*
in.the-mornings stand.M.SG the-robot-M.PL in-line

2.2 Colloquial Hebrew

Preminger (2009; 2014), the only generative study of Lack of ϕ-agreement in Hebrew VS, puts forth a framework in which sentences involving attempted-but-failed agreement are nonetheless grammatical. He treats the lack of agreement in VS order in Hebrew as a case of agreement failure, not agreement optionality, despite appearances (9a-9b). According to him, this ϕ-agreement failure is tolerated only in the presence of an intervener bearing ϕ-features between the verb and its postverbal subject – specifically, a possessive dative; it is impossible otherwise (9c).

(9) a. *nafl-a le-dani ha-cincenet.*
fell-3F.SG DAT-Dani the-jar.F.SG
'Dani's jar fell.'

b. *?nafal le-dani ha-cincenet.*
fell.3M.SG DAT-Dani the-jar.F.SG (Preminger, 2009, 242:(2))

c. **nafal ha-cincenet.*
fell.3M.SG the-jar.F.SG (Preminger, 2009, 244:(7b))

Preminger locates the possessive dative in the Specifier position of an Applicative Phrase, where it is within the minimal search domain of T, but not that of the internal subject (the possessee) because the possessive dative c-commands the subject (possessee) but not vice versa (10). Hence, the possessive dative (PD) intervenes in the relation between T and the postverbal subject, thereby permitting ϕ-agreement failure. The failure does not involve agreement with the dative intervener, as shown in (11). [3]

'The chairs are standing here.'

b. *akca oti dvora.*
stung.3F.SG ACC.me bee.F.SG
'A bee stung me.' (Melnik, 2002, 19:(35))

(ii) a. **omed po ha-kisa'ot.*
stand.M.SG here the-chair.M.PL

b. **akac oti dvora.*
stung.3M.SG ACC.me bee.F.SG

[3] Providing examples with passives of ditransitive verbs taking an argumental dative (i-ii), Preminger

(10) $[_{TP}$ T $[_{vP}$ $[_{ApplP}$ PD $[_{VP}$ $[_{DP}$ Possessee]]]]]

(11) **nafla le-dina ha-maftexot.*
fell.3F.SG DAT-Dina(F) the-key.M.PL
'Dina's keys fell.'

To explain why ϕ-agreement is nonetheless possible despite the intervention (9a), Preminger suggests that in these cases the postverbal subject covertly moves up to a position lower than T, but higher than the possessive dative, where the latter no longer intervenes. Hence, ϕ-agreement is possible. This movement has no overt parallel and is not further motivated by Preminger.

More importantly, there are fairly frequent examples from everyday speech, where the unaccusative verb and its internal argument fail to establish a ϕ-agreement relation even in the absence of an intervening possessive dative, as illustrated by examples (1a-1c) above and (12a-12c) below. The existence of such examples casts doubt on Preminger's claim.

(12) a. *nigmar ha-hastara.*
finished.3M.SG the-hiding.F.SG
'No more hiding.' (`https://tinyurl.com/25ajk9da`, 6 min 11 s)

b. *matxil ha-nisu'im nigmar ha-ahava.*
start.3M.SG the-marriage.M.PL end.3M.SG the-love.F.SG
'When marriage begins, love ends.' (Kuzar, 2002)

c. *lo nišar mayim xam-im.*
NEG remained.3M.SG water.M.PL hot-M.PL
'There is no more hot water.' (`https://tinyurl.com/y9fer3zv`)

Further, examples where an adverbial lacking accessible ϕ-features intervenes between the verb and its postverbal subject also exist (13a-13b).

(13) a. *bo hofia larišona ha-mila 'robot'*
in.him appeared.3M.SG for.the.first.time the-word.F.SG 'robot'
'There the word 'robot' appeared for the first time.'
(`https://tinyurl.com/4xk4subw`)

argues that an argumental dative does not license ϕ-agreement failure. It does not constitute an intervener between T and the direct object, as both of them are in the same minimal domain.

(i) *nimsera la-mefake'ax ha-ma'atafa.*
handed.PASS.3F.SG DAT.the-supervisor.M.SG the-envelope.F.SG
'The envelope was handed to the supervisor.'

(ii) **nimsar la-mefake'ax ha-ma'atafa.*
handed.PASS.3M.SG DAT.the-supervisor.M.SG the-envelope.F.SG

b. *ad še-higia pitom averat bam.*
until that-arrived.3M.SG suddenly violation.CS.F.SG information.security
'Until an information security violation was suddenly discovered.'
(`https://tinyurl.com/48dnnxcj`)

Finally, there are examples with a quantified postverbal subject (14a-14b). The quantifier does not intervene between the verb and the subject as it is part of the noun phrase subject.

(14) a. *nišpax harbe mayim.*
spilled.3M.SG lots.of water.M.PL
'A lot of water was spilled.' (`https://tinyurl.com/4jj4fdrv`)

b. *nocar hamon balagan-im.*
created.PASS.3M.SG lots.of mess-M.PL
'A big mess was created.' (`https://tinyurl.com/5crp3u42`, 5 min 30 s)

Clearly, there is a discrepancy between the occurrence of examples such as (12-14) as well as (1) in corpora and Preminger's observation regarding the crucial role of intervention. Since the empirical domain of investigation belongs to a low-register variety of Hebrew and given the recency of the phenomenon, authors' intuitions and attested examples are insufficient to obtain a solid empirical basis. We therefore conducted two experiments to shed light on the issue of intervention. Experiment 1 examined the acceptability of VS sentences lacking ϕ-agreement between the verb and its subject, both with and without an intervening possessive dative. Experiment 2 compared three types of interveners – possessive datives, adverbs and quantifiers – to sentences without intervention.

Before turning to the experiments, a few words on grammaticality and acceptability are in order. Following Keller (2000), we assume that grammaticality is binary, such that sentences are either grammatical or ungrammatical. Acceptability, by contrast, is graded, meaning that both grammatical and ungrammatical sentences may be associated with varying degrees of acceptability. The factors that are responsible for this variance are presumed to be extra-grammatical, e.g., plausibility, frequency, processing ease, etc. (for discussions, see Sorace and Keller (2005); Sprouse (2007), among others). Since the present experiments examine a phenomenon characteristic of low-register varieties of Hebrew and one that contradicts both the prescriptive rule and actual usage of standard Hebrew, high acceptability ratings for sentences lacking ϕ-agreement are not expected. Nonetheless, significant differences between the different conditions could shed light on the data.

3 Experiment 1

Experiment 1 examined the acceptability of sentences where the verb is singular and the subject plural (henceforth: Singular-Plural) comparing them to two baselines: sentences where both the verb and the subject are plural (grammatical baseline), and sentences where the verb is plural and the subject singular (ungrammatical baseline). Each of these three conditions was tested with and without the intervention of a possessive dative.

The predictions were as follows. The Agreement condition was predicted to be acceptable across the board and significantly more acceptable than the other two conditions given its normative status and occurrence in all registers of Hebrew. The Plural-Singular condition (the ungrammatical baseline) was predicted to be consistently unacceptable. Predictions regarding the Singular-Plural condition diverged. If Preminger is right and lack of ϕ-agreement is licensed only by intervention, specifically, by a possessive dative intervener, then Singular-Plural sentences should be significantly more acceptable than the ungrammatical baseline only in the presence of such intervention. In contrast, based on attested examples, Singular-Plural sentences are expected to be significantly more acceptable than the ungrammatical baseline irrespective of intervention. Nevertheless, due to their substandard status, they are not expected to receive high acceptability ratings.

3.1 Method

3.1.1 Participants

42 adults aged 22-50 (mean age = 30.45) participated in the study. All participants were monolingual native speakers of Hebrew with no background in linguistics.

3.1.2 Materials

The experiment tested the acceptability of 3 types of ϕ-relations between the verb and its post-verbal subject: plural agreement (Agreement), lack of agreement in which the verb was singular but the subject plural (Singular-Plural), and the reverse – verb plural and subject singular (Plural-Singular). These ϕ-relation types were crossed with the Intervention variable: the presence vs. absence of a possessive dative intervener between the verb and its internal argument (with intervention vs. without intervention). This design yielded a total of 6 conditions: Agreement without intervention (15a), Agreement with intervention (15b), Singular-Plural without intervention (15c), Singular-Plural with intervention (15d), Plural-Singular without intervention (15e), and Plural-Singular with intervention (15f).

(15) a. *naflu ha-maftexot ba-sedek.*
fell.3PL the-key.M.PL in.the-crack
'The keys fell in the crack.'

b. *naflu le-dan ha-maftexot ba-sedek.*
fell.3PL DAT-Dan the-key.M.PL in.the-crack
'Dan's keys fell in the crack.'

c. *nafal ha-maftexot ba-sedek.*
fell.3M.SG the-key.M.PL in.the-crack

d. *nafal le-dan ha-maftexot ba-sedek.*
fell.3M.SG DAT-Dan the-key.M.PL in.the-crack

e. **naflu ha-mafte'ax ba-sedek.*
fell.3PL the-key.M.SG in.the-crack

f. **naflu le-dan ha-mafte'ax ba-sedek.*
fell.3PL DAT-Dan the-key.M.SG in.the-crack

The unaccusative verbs used in the experiment were selected based on the following diagnostics:

(16) The characteristics of unaccusative verbs:

1. The verb occurs in the VS word order, which is typical of verbs whose subject is an internal argument (Shlonsky, 1997; Siloni, 2012).
2. The verb licenses a possessive dative (Borer and Grodzinsky, 1986; Meltzer-Asscher and Siloni, 2012; Plotnik et al., 2024).
3. The verb has a transitive counterpart whose external argument has a Cause role (i.e. an argument unspecified regarding mental state) (Reinhart, 2002).

Two passives were included in the experiment. [4]

18 sets of experimental items, each comprising 6 conditions were constructed. Items were distributed across 6 lists in a Latin-square design, and participants were evenly assigned to lists. Each list included a total of 54 sentences: 18 experimental items and 36 filler items. The filler items differed from the experimental items in their syntactic structure, and spanned various levels of acceptability. Specifically, they included 12 grammatical sentences (e.g. *dibarti im dina etmol* 'I talked to Dina yesterday'), 12 somewhat marginal sentences (due to the adverb position, e.g. *?hašaršeret be-pit'omiut nigneva etmol* 'The necklace suddenly was stolen yesterday') and 12 ungrammatical sentences (due to their null subject in third

[4] These instances, namely the passives of *sagar* 'close' and *ganav* 'stole', are attested in colloquial Hebrew. More generally, the Hebrew passive is mostly found in high registers, where lack of ϕ-agreement between verb and subject is disallowed. Hence, passive verbs were not included in the following experiment.

person, which is impossible in Hebrew. E.g., **taram le-david sfarim be-yanuar* 'Donated books to David in January'). Each participant saw one experimental sentence of each set, thus encountering 3 sentences per condition. The sentences were presented in a pseudo-randomized order, such that 2 filler items appeared between any two experimental items. The experiment was conducted using the Ibex Farm web platform (Zehr and Schwarz, 2018).

3.1.3 Procedure

A link to the experiment was posted on social media and sent individually to potential participants. Participants were first asked to fill out demographic information and provide informed consent. They were instructed to rate the sentences to be presented on the screen on a scale of 1 ("unacceptable") to 7 ("acceptable"), ignoring prescriptive Hebrew rules taught in school. Instead, they were asked to judge whether each sentence sounds acceptable (natural) in everyday speech according to their intuitions as native speakers. An example of an acceptable sentence (*ra'iti xatul lavan* 'I saw a white cat') and an unacceptable sentence (**ra'iti lavan xatul* 'I saw a cat white') was provided. After a single grammatical practice sentence, the experiment began.

3.2 Results

Table 1: Fixed effects estimates from a linear mixed model analyzing acceptability ratings by Agreement Type, Intervention, and their interaction. Random intercepts for participants and for items (nested within set) were included. Reference levels: Agreement = Agreement, Intervention = No Intervention.

	Estimate	Std. Error	df	t	p
Intercept	5.87	0.16	113.05	36.03	< 0.001***
Agreement:Singular-Plural	-3.83	0.18	97.55	-21.09	< 0.001***
Agreement:Plural-Singular	-4.36	0.18	97.55	-24.02	< 0.001***
Intervention:Intervention	0.13	0.18	97.55	0.74	0.46
Agreement:Singular-Plural:Intervention:Intervention	0.68	0.26	97.55	2.67	0.01**
Agreement:Plural-Singular:Intervention:Intervention	-0.02	0.26	97.55	-0.06	0.95

A linear mixed model fit by REML using Satterthwaite's method revealed a significant main effect for agreement type. Specifically, both Singular-Plural (estimate = -3.83, t(97.55) = 21.09, $p < 0.001$) and Plural-Singular (estimate = -4.36, t(97.55) = 24.02, $p < 0.001$) significantly reduced the acceptability compared to the grammatical baseline. The effect of intervention varied by agreement type. A significant interaction was found between Singular-Plural and Intervention (estimate = 0.68, t(97.55) = 2.67, $p < 0.001$). In contrast, the

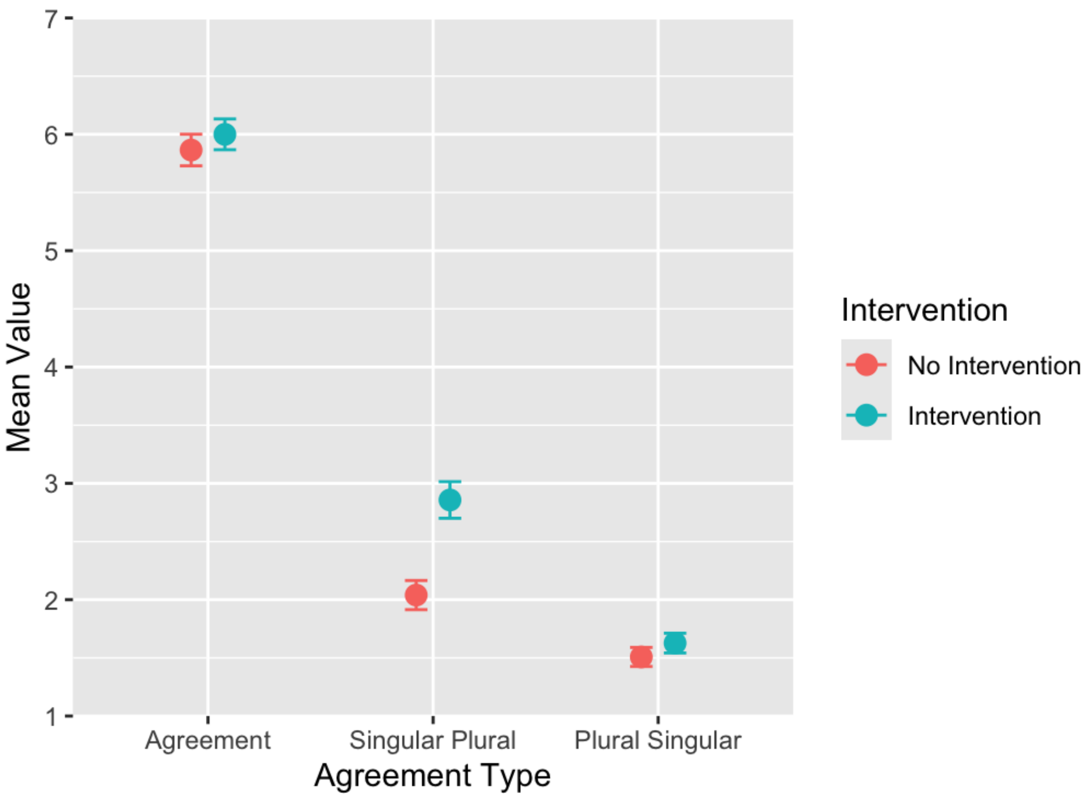


Figure 1. Mean ratings (± standard error) of acceptability across Agreement Types grouped by Intervention. Agreement: No Intervention: M = 5.87, SD = 1.53; Intervention: M = 6, SD = 1.49; Singular–Plural: No Intervention: M = 2.04, SD = 1.41; Intervention: M = 2.86, SD = 1.77; Plural–Singular: No Intervention: M = 1.51, SD = 0.92; Intervention: M = 1.63, SD = 0.95.

interaction between Plural-Singular and Intervention was not significant (estimate = 0.02, t(97.55) = 0.06, p = 0.95), as well as the interaction between Agreement and Intervention (estimate = 0.13, t(97.55) = 0.74, p = 0.46). Posthoc contrasts were examined to further explore the interaction effects. The difference between Singular-Plural and Plural-Singular was more pronounced with intervention (estimate = 1.230, t(97.5) = 6.78, $p < 0.001$) than in its absence (estimate = 0.532, t(97.5) = 2.93, p = 0.02), yet a significant difference was found in both cases.

3.3 Discussion

Agreement sentences were found to be significantly more acceptable than Singular-Plural and Plural-Singular sentences. Singular-Plural sentences were found to be significantly more acceptable than Plural-Singular sentences. Intervention significantly improved the acceptability of the Singular-Plural condition, but not that of the Agreement or Plural-Singular conditions. In other words, intervention affects only the acceptability of the Singular-Plural condition, with no impact on the grammatical and ungrammatical baselines. The difference between Singular-Plural and Plural-Singular agreement sentences was more pronounced with intervention than in its absence, though a significant difference was found in both cases.

Thus, as argued by Preminger, the intervention of a possessive dative clearly improves lack of ϕ-agreement (i.e. the Singular-Plural condition). However, the ranking of Singular-Plural sentences was lower than expected not only without intervention (M = 2.04) but also with intervention (M = 2.85), results that contrast with the presence of both types in colloquial speech and informal writing. These low rankings call for an explanation, regardless of intervention. First, as already mentioned, Singular-Plural sentences are limited to low-register varieties and constitute a rather new linguistic development that contradicts both the prescriptive rule and actual usage of standard Hebrew. Despite instructions to disregard prescriptive grammar rules, participants were likely influenced by the normative expectation that verbs agree with their subjects. Moreover, the Singular-Plural sentences were tested in comparison to Agreement sentences, which are grammatical and prevalent across all registers of Hebrew, both high and low, and whose high scores in both conditions (Intervention: M = 6.00, No Intervention: M = 5.68) may have lowered the ranking of the other conditions. Plural-Singular sentences are unattested and ungrammatical, and indeed, despite the low average of Singular-Plural sentences, they were judged as significantly more acceptable than the Plural-Singular condition.[5]

[5] A post-hoc analysis indicates that the distribution across participants is not bimodal, but two groups can nevertheless be identified. However, the sample sizes of the two groups are too small to yield reliable statistical results. The grammatical fillers averaged 6.83, the marginal fillers 3.53, and the ungrammatical

To better understand the status of Singular-Plural sentences and the role of intervention, we conducted Experiment 2, which tested the acceptability of Singular-Plural sentences with additional intervening elements, such as adverbs and quantifiers. These sentences were compared to Singular-Plural sentences with possessive dative intervention and to Singular-Plural sentences without intervention. To control for confounding factors and prevent carryover effects, the experiment employed a between-group design in which a control group was exposed to the same sentences but with the grammatical baseline (Agreement) instead of the Singular-Plural condition. Finally, to allow comparison of the Singular-Plural condition with constructions of the same low register, fillers consisted mostly of colloquial structures unacceptable in standard Hebrew.

4 Experiment 2

Experiment 2 tested the acceptability of sentences with three types of interveners: possessive datives, adverbs and quantifiers. These sentences were compared to sentences without intervention. A between-group design was used: participants were divided into two groups with the first group exposed to the four types of intervention within Singular-Plural sentences, and the second group exposed to the same conditions within Agreement sentences. The second group served as a control group. Eliminating the influence of the agreement factor allowed us to control for differences in acceptability judgments that might result from confounding factors or carryover effects rather than from the absence of agreement itself.

Since quantifiers are dominated by the postverbal subject DP, they do not constitute structural intervention, on a par with the no-intervention condition. Adverbs, being DP-external and possibly outside the VP minimal domain, do not, unlike possessive datives, have accessible ϕ-features. Accordingly, under Preminger's approach, ϕ-agreement failure between T and the subject DP is expected only in the possessive dative condition. Preminger's approach therefore predicts that the possessive dative condition should receive higher acceptability ratings than all other conditions.

By contrast, drawing on attested examples and the results of Experiment 1, we expected that the possessive dative condition would be rated higher than the no-intervention conditions (including the quantifier condition), but not higher than the adverb condition, the latter and the possessive dative condition constituting structural interveners. Additionally,

ones 3.11. The marginal fillers involved slight syntactic anomaly as the adverb did not appear in its most suitable position. The ungrammatical fillers contained a null third-person subject, which is ungrammatical in Hebrew matrix clauses. Nonetheless, they received relatively high ratings, likely because they are acceptable fragments (with nothing preceding the verb). Since the instructions did not explicitly require full sentences, participants may have interpreted them as acceptable sentence chunks.

we expected the ratings of the Singular–Plural conditions to be higher than in Experiment 1.

4.1 Method

4.1.1 Participants

The first group consisted of 104 adults aged 20-39 (mean age = 27.78) and the second group consisted of 104 adults aged 18-39 (mean age = 26.36). All participants were monolingual native speakers of Hebrew with no background in linguistics.

4.1.2 Materials

The materials for the first group included the four different intervention conditions – illustrated below – presented in Singular-Plural sentences, whereas those for the second group involved the same conditions in Agreement sentences. The four conditions were as follows: possessive dative intervention (17a), quantifier intervention (17b), adverb intervention (17c), and no intervention (17d). [6]

(17) a. *nidbak/nidbeku le-maya alim la-šimša.*
stuck.3M.SG/3PL DAT-Maya leaf.M.PL to.the-windowpane
'Maya had leaves stuck to the windowpane.'

b. *nidbak/nidbeku hamon alim la-šimša.*
stuck.3M.SG/3PL lots.of leaf.M.PL to.the-windowpane
'Lots of leaves stuck to the windowpane.'

c. *nidbak/nidbeku šuv alim la-šimša.*
stuck.3M.SG/3PL again leaf.M.PL to.the-windowpane
'Leaves stuck to the windowpane again.'

d. *nidbak/nidbeku alim la-šimša.*
stuck.3M.SG/3PL leaf.M.PL to.the-windowpane
'Leaves stuck to the windowpane.'

All verbs were unaccusative verbs, classified as such based on the characteristics outlined in (16) above. The quantifiers were of the type that could not trigger singular agreement

[6] In Experiment 1, we used examples with definite postverbal subjects. This choice reflects their high frequency in attested cases of non-agreeing VS order, especially when nothing intervenes between the verb and the subject. In Experiment 2, we used indefinite subjects to allow for the inclusion of the quantifier condition.

with the verb when modifying a plural head noun (see Danon (2013)).[7] As for the internal arguments, none of them had a mass reading that licenses singular agreement.

There were 12 sets of experimental items, each consisting of 4 conditions. The items were divided into 4 lists in a Latin-square design, and participants were evenly assigned to lists. Each list contained a total of 36 sentences: 12 experimental items and 24 fillers. The fillers for the Singular-Plural group consisted of 6 sets of 4 sentences each. The sets included structures resulting from linguistic changes at different periods of time. One set consisted of sentences that began emerging during the revival of Hebrew, and are now grammatical (*haya po et...* 'There was here ACC'; see 5.4). Another set comprised colloquial structures (discussed by Ariel (1999)) involving the dropping of the complex [preposition-resumptive pronoun] in relative clauses, which are disallowed in standard Hebrew. Four additional sets reflected more recent changes, all yielding colloquial structures unacceptable in standard Hebrew: two sets involved an overt expletive in contexts where it is disallowed in the standard language (e.g. *ze kar be-london* 'it is cold in London'; *ze efšar la-azov* 'it is possible to leave'), one set showing lack of agreement between the modifier *same* and the noun, and one set involving the use of the preposition *al* 'on' instead of the verb-selected preposition.

The fillers for the Agreement group consisted of 6 sets of 4 sentences each: 3 sets of grammatical sentences and 3 sets of ungrammatical sentences. Each participant was presented with one experimental item of each set, thus being exposed to 3 sentences of each condition. The sentences were presented in a pseudo-randomized order, such that 2 filler items appeared between any two experimental items. The experiment was conducted using the Ibex Farm web platform (Zehr and Schwarz, 2018).

4.1.3 Procedure

The procedure was identical to that of Experiment 1.

4.2 Results

A linear mixed model fit by REML using Satterthwaite's method found a significant main effect for agreement (estimate = 3.46, t(2383.25) = 30.59, $p < 0.001$). As shown in Table 2 and in Figure 2, within the Singular-Plural group, the possessive dative condition was significantly more acceptable than all other conditions: the adverb condition (estimate = -0.84, t(69.5) = 6.25, $p < 0.001$), the quantifier condition (estimate = -0.9, t(69.5) = 6.68, $p < 0.001$), and no intervention (estimate = -1.05, t(69.5) = 7.83, $p < 0.001$). Other

[7] The quantifiers were equally divided to quantifiers that are limited to count nouns, and quantifiers that are neutral in terms of countability. The adverbs were also equally divided to adverbs of manner and adverbs of time.

Table 2: Fixed effects estimates from a linear mixed model analyzing acceptability ratings by Agreement Type, Condition, and their interaction. Random intercepts for participants and for items (nested within set) were included. Reference levels: Agreement = Singular-Plural, Condition = Possessive Dative.

	Estimate	Std. Error	df	t	*p*
Intercept	3.86	0.13	89.39	28.94	< 0.001***
Agreement:Agreement	2.3	0.11	2383.25	20.36	< 0.001***
Condition:No Intervention	-1.05	0.13	68.33	-7.83	< 0.001***
Condition:Quantifier	-0.9	0.13	68.33	-6.68	< 0.001***
Condition:Adverb	-0.84	0.13	68.33	-6.25	< 0.001***
Agreement:Agreement:Condition:No Intervention	1.16	0.16	2310.34	7.46	< 0.001***
Agreement:Agreement:Condition:Quantifier	0.83	0.16	2310.34	5.33	< 0.001***
Agreement:Agreement:Condition:Adverb	0.21	0.16	2310.34	1.36	0.173

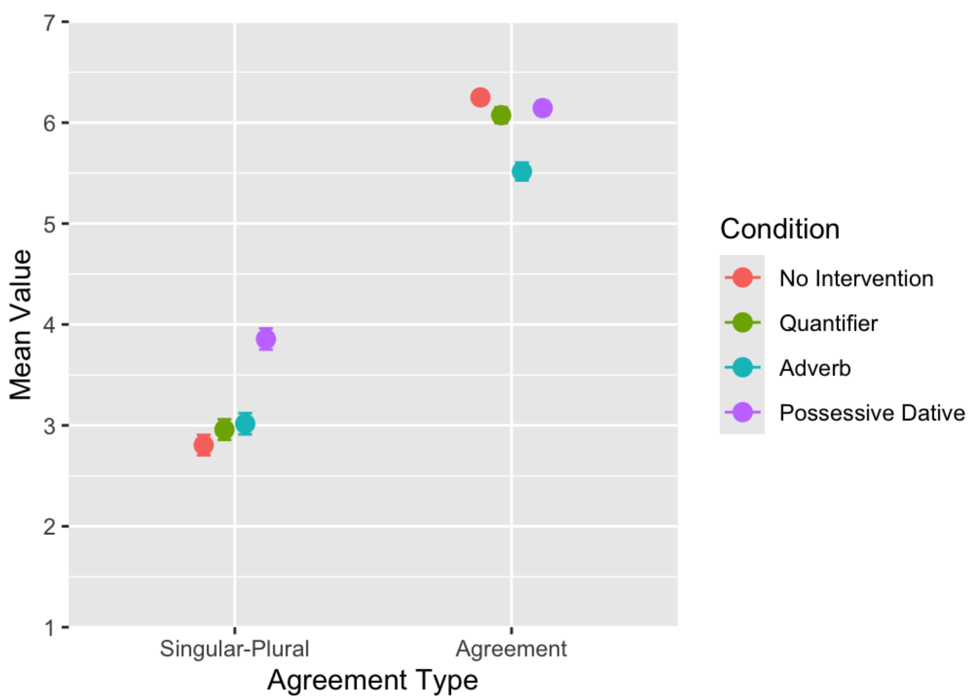


Figure 2. Mean ratings ($\pm$ standard error) of acceptability across Agreement Type grouped by Condition. Singular-Plural: No Intervention: M = 2.8, SD = 1.77; Quantifier: M = 2.96, SD = 1.79; Adverb: M = 3.02, SD = 1.85; Possessive Dative: M = 3.86, SD = 1.83; Agreement: No Intervention: M = 6.25, SD = 1.14; Quantifier: M = 6.07, SD = 1.38; Adverb: M = 5.52, SD = 1.55; Possessive Dative: M = 6.14, SD = 1.22.

comparisons were non-significant (No Intervention-Quantifier: estimate = 0.15, t(69.5) = 1.15, $p = 0.66$; No Intervention-Adverb: estimate = 0.21, t(69.5) = 1.58, $p = 0.4$; Quantifier-Adverb: estimate = 0.06, t(69.5) = 0.43, $p = 0.97$). Within the Agreement group, the adverb condition was significantly less acceptable than all other conditions: the possessive dative condition (estimate = 0.63, t(69.5) = 4.68, $p < 0.001$), the quantifier condition (estimate = 0.56, t(69.5) = 4.15, $p < 0.001$), and no intervention (estimate = 0.73, t(69.5) = 5.46, $p < 0.001$). Other comparisons were non-significant (Possessive Dative-No Intervention: estimate = 0.11, t(69.5) = 0.79, $p = 0.86$; Possessive Dative-Quantifier: estimate = -0.07, t(69.5) = 0.53, $p = 0.95$; No Intervention-Quantifier: estimate = -0.18, t(69.5) = 1.31, $p = 0.56$).

A significant interaction was found between Agreement type (Agreement vs. Singular-Plural) and possessive dative vs. quantifier (estimate = -0.83, t(1116) = 5.26, $p < 0.001$) as well as between Agreement type (Agreement vs. Singular-Plural) and possessive dative vs. no intervention (estimate = -1.16, t(1116) = 7.53, $p < 0.001$). However, as shown in Figure 3, there was no significant interaction between Agreement type (Agreement vs. Singular-Plural) and possessive dative vs. adverb (estimate = -0.21, t(1116) = 1.28, $p = 0.2$).

Since the ranking of Singular-Plural sentences was still relatively low, we conducted post-hoc analyses to examine potential subgroups of items or participants in our data.[8] Silverman's test found no evidence of bimodality in the distribution of acceptability ratings at the participant ($p = 0.73$), trial ($p = 0.13$), or item-by-condition level. A two-component Gaussian mixture model fit the data significantly better than a one-component model based on AIC (ΔAIC = 33.04), yet BIC showed a smaller difference, preferring a one-component model (ΔBIC = 6.62). The division into two clusters revealed a variation in participant response patterns, suggesting the presence of a *conservative* group (n = 44; M = 2.05) and an *innovator* group (n = 60; M = 3.97) in our study. As shown in Figure 4, the conservatives, represented by Cluster 1, ranked Singular-Plural sentences significantly lower than the innovators, represented by Cluster 2 (estimate = 1.92, t(102) = 11.83, $p < 0.001$). No correlation with age was found ($p = 0.06$).

Examining the innovator Cluster 2 alone, the possessive dative condition was again significantly more acceptable than all other conditions: the adverb condition (estimate = -1.02, t(177) = 5.93, $p < 0.001$), the quantifier condition (estimate = -1.06, t(177) = 6.15, $p < 0.001$), and no intervention (estimate = -1.22, t(177) = 7.09, $p < 0.001$). Other comparisons were non-significant (Adverb-No Intervention: estimate = -0.2, t(177) = 1.17, $p = 0.65$; Adverb-Quantifier: estimate = -0.04, t(177) = 0.23, $p = 0.99$; Quantifier-No Intervention: estimate = -0.16, t(177) = 0.94, $p = 0.78$). Likewise, there was no significant interaction

[8] We thank an anonymous reviewer for this suggestion.

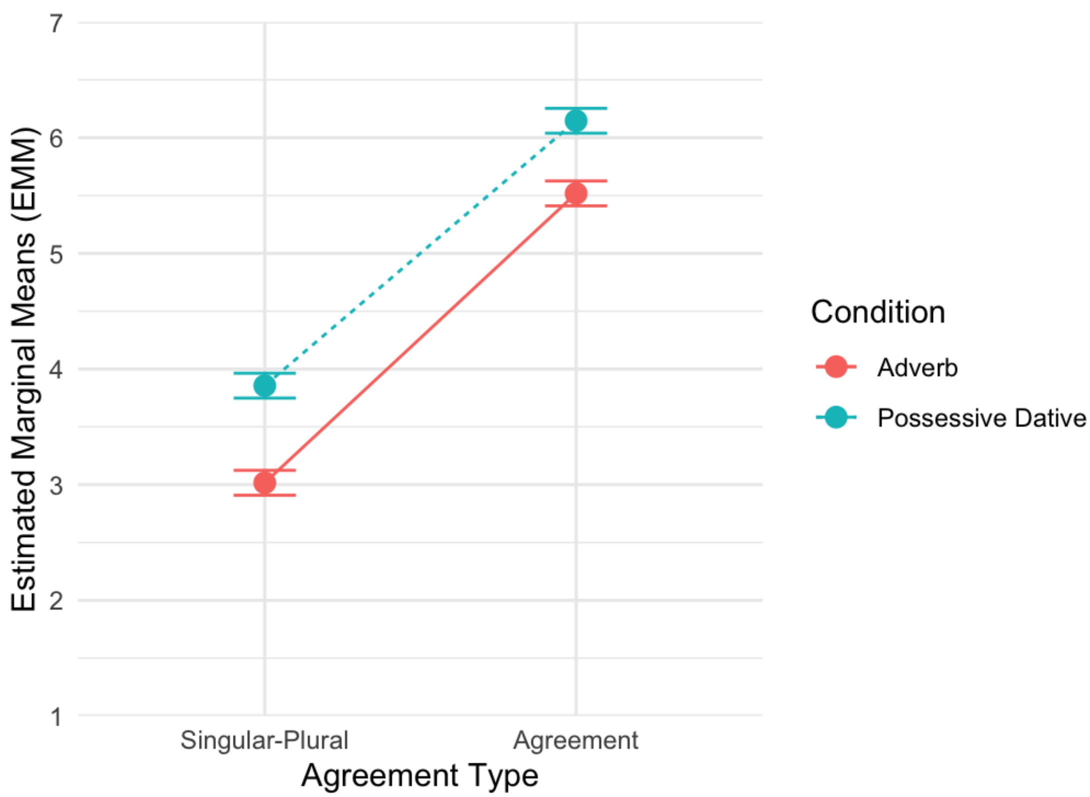


Figure 3. Interaction plot showing estimated marginal means by Agreement Type (Agreement vs. Singular-Plural) and Condition (Adverb vs. Possessive Dative).

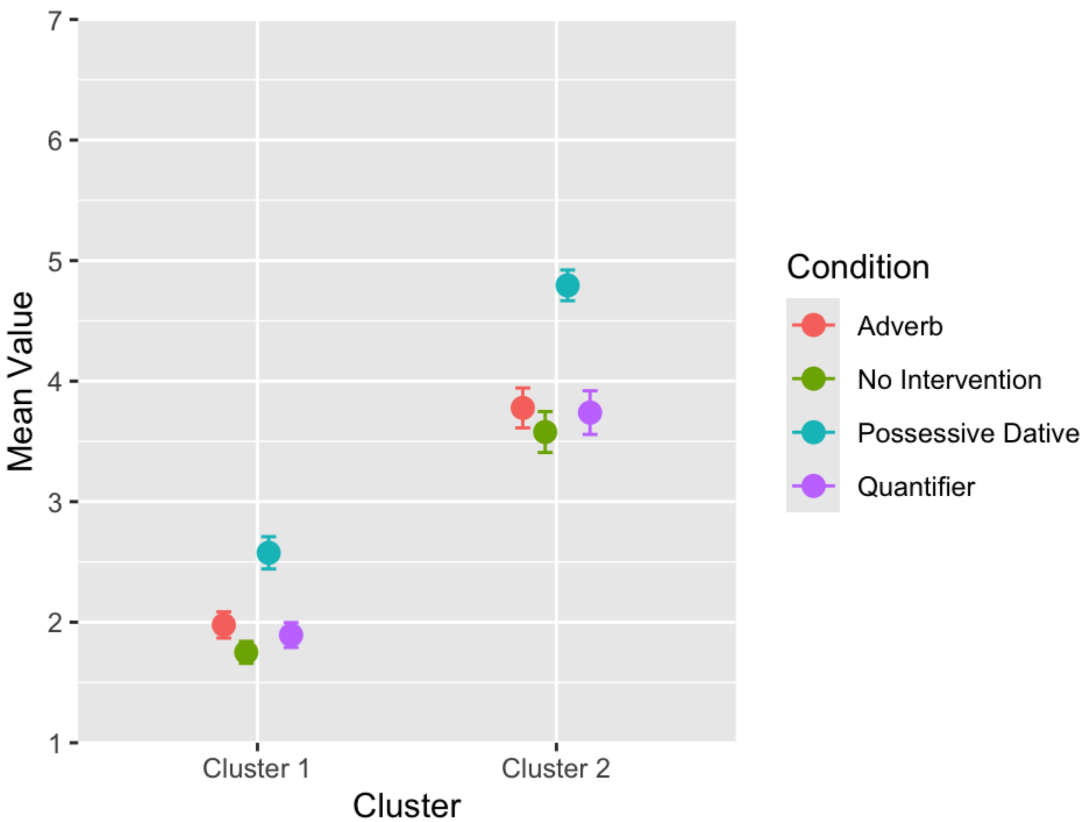


Figure 4. Mean ratings (± standard error) for each condition by mixture model cluster. Cluster 1: Possessive Dative: M = 2.58, SD = 0.89; Adverb: M = 1.98, SD = 0.72; Quantifier: M = 1.89, SD = 0.69; No Intervention: M = 1.75, SD = 0.61. Cluster 2: Possessive Dative: M = 4.79, SD = 0.99; Adverb: M = 3.78, SD = 1.28; Quantifier: M = 3.74, SD = 1.4; No Intervention: M = 3.58, SD = 1.31.

between Agreement type (Agreement vs. Singular-Plural) and possessive dative vs. adverb ($F(1, 322) = 2.26$, $p = 0.13$).[9]

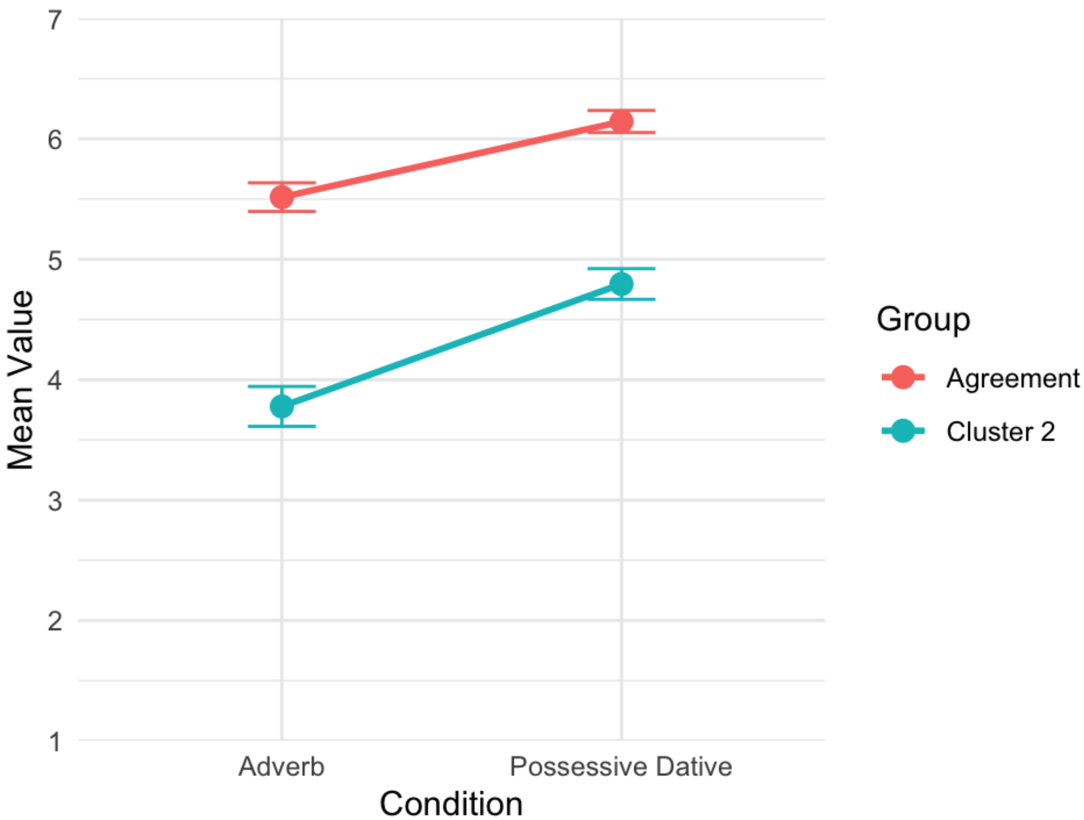


Figure 5. Interaction plot showing estimated marginal means by group (Agreement vs. Singular-Plural Cluster 2) and Condition (Adverb vs. Possessive Dative).

We also examined the ranking of the colloquial fillers in the Singular-Plural group. Their mean rating in the entire group was 3.4.[10] In the innovator Cluster 2 there was no significant difference between the acceptability ratings of the experimental items and those of the colloquial fillers (estimate = -0.1, t(516) = 0.65, $p = 0.52$). [11] The comparison of the experimental conditions, excluding the possessive dative, with the colloquial fillers was also

[9] For completeness, the conservative Cluster 1 yielded the same pattern of results: the possessive dative condition was significantly more acceptable than all other conditions (Adverb: estimate = -0.6, t(129) = 5.38, $p < 0.001$; Quantifier: estimate = -0.68, t(129) = 6.13, $p < 0.001$; No Intervention: estimate = -0.83, t(129) = 7.42, $p < 0.001$), and other pairwise comparisons were non-significant (Adverb-No Intervention: estimate = -0.23, t(129) = 2.04, $p = 0.18$; Adverb-Quantifier: estimate = -0.08, t(129) = 0.75, $p = 0.88$; Quantifier-No Intervention: estimate = -0.14, t(129) = 1.29, $p = 0.57$). Finally, there was no significant interaction between Agreement type (Agreement vs. Singular-Plural) and possessive dative vs. adverb ($F(1, 290) = 0.01$, $p = 0.91$).

[10] For completeness, in the Agreement group, the grammatical fillers averaged 6.7, and the ungrammatical ones 1.59.

[11] The mean rating of the grammatical filler (*haya po et* 'There was here ACC'), which (being grammatical)

non-significant. In contrast, in the conservative Cluster 1 the rating of the experimental items was significantly lower than that of the fillers (estimate = 0.72, t(516) = 3.93, $p < 0.001$).

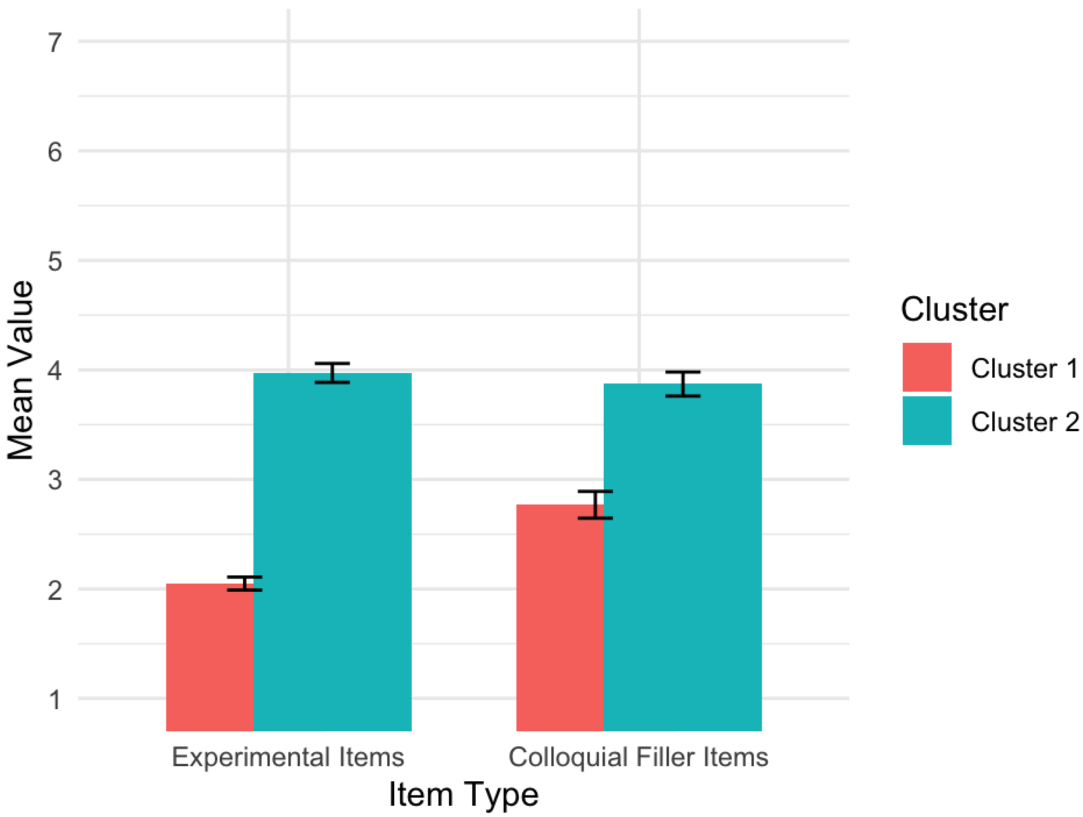


Figure 6. Mean ratings (± standard error) for experimental and colloquial filler items by cluster. Cluster 1: Experimental Items: M = 2.05, SD = 0.79; Filler Items: M = 2.77, SD = 0.81. Cluster 2: Experimental Items: M = 3.97, SD = 1.34; Filler Items: M = 3.87, SD = 0.85.

4.3 Discussion

In both the entire Singular-Plural group and the innovator cluster, the same statistical patterns were observed. However, whereas the overall group rated Singular-Plural sentences relatively low, in the innovator cluster they were as acceptable as the colloquial fillers (Figure 6). Within the Singular-Plural sentence type in both groups, the possessive dative condition was found to be significantly more acceptable than all other conditions. No other significant

was not part of the calculation, was 5.64. The [preposition-resumptive pronoun] dropping fillers received an average of 4.27, and all other fillers received a rating of around 3 (weather expletives 3.21, other expletives 3.1, modifier-noun disagreement 3.06, and overuse of al 'on' 3.37), with an average of 3.19.

differences were observed. This, in itself, could be seen as reinforcing Preminger's claim that it is the possessive dative that licenses Singular-Plural sentences. However, the quantifier and no-intervention conditions were significantly less acceptable than the possessive dative in the Singular-Plural type, but *not* in the Agreement type. In contrast, the adverb condition was rated lower than the possessive dative overall, across both sentence types, Agreement and Singular-Plural sentences. As shown in Figures 3 and 5, there was no significant interaction between sentence type (Agreement vs. Singular-Plural), and the contrast between the adverb and the possessive dative. That is, there was no significant difference in how the adverb and the possessive dative conditions affected the Agreement sentences and the Singular-Plural sentences. This means that the lower ratings for the adverb condition in the Singular-Plural sentence type are not due to lack of agreement, since similarly low ratings were observed in the Agreement sentence type as well. For ease of discussion, we refer to this consistent drop in ratings caused by the adverb as *the adverb decrease*. This decrease appears to be independent of agreement and should stem from a separate factor. If this decrease were neutralized, there would be no significant difference between the adverb and the possessive dative in either sentence type. The fact that the adverb did not lead to a greater decrease in the Singular-Plural type compared to the Agreement type implies that both the possessive dative and the adverb facilitate the absence of agreement. This contrasts with the quantifier and no-intervention conditions, which do not show this facilitating effect.

This pattern challenges Preminger's "failure to agree" analysis, as adverbs lack accessible ϕ-features and therefore do not interfere with the ϕ-relations between the verb and its postverbal subject. Moreover, we interpret the impact of the intervention as a facilitating one; that is, the possessive dative and the adverb improve acceptability rather than constituting a prerequisite for grammaticality (contra Preminger). This is supported by the fact that the acceptability ratings of the experimental conditions – even when the higher-rated possessive dative is excluded – do not differ significantly from those of the colloquial fillers.

The quantifier, although constituting a linear intervener between the verb and the head noun, is part of the noun phrase and may itself signal the plurality of the noun; it is therefore not surprising that it patterns with the no-intervention condition.

In sum, not only the possessive dative but also the adverb exerts a similar facilitating effect, unlike the no-intervention conditions (including the quantifier condition). We argue that Preminger's "failure-to-agree" analysis of the construction is untenable. That is, this construction does not support his theoretical account of grammaticality, in which sentences involving attempted-but-failed agreement can nonetheless be deemed grammatical. In section 5.3 we offer an alternative account of the facilitating effect of intervention. To further investigate the source of the "adverb decrease", we conducted an additional experiment, the details of which are provided in the appendix.

5 Proposal: Agreement and case

We argue that the loss of ϕ-agreement on the postverbal subject in Hebrew results from its analysis as caseless. More generally, we claim that unaccusative constructions involving a postverbal subject exhibit instability in terms of case and consequently agreement. Specifically, in languages like Hebrew that do not morphologically mark nominative, the internal argument subject in its merger position (i.e. complement position) may be analyzed by speakers as caseless. Section 5.1 provides some crosslinguistic evidence for the instability of agreement in VS configurations that are unmarked for nominative. Section 5.2 discusses the loss of nominative in Hebrew and presents corroborating evidence. The process leading from the loss of nominative to the loss of agreement is discussed in Section 5.3. Section 5.4 is devoted to Hebrew existential and possessive constructions, showing that they have undergone a parallel process in the past century. Finally, Section 5.5 briefly addresses Dependent Case Theory.

5.1 Unstable configuration

Crosslinguistic variation lends some support to our claim that the agreement patterns with a postverbal internal argument subject are indeed unstable. In English existential constructions, the verb regularly agrees with its postverbal subject.

(18) a. There are lots of cars in the car park.
     b. There are books on the desk.

Nevertheless, in colloquial English, lack of agreement is also attested in such constructions:

(19) a. There's lots of cars in the car park. (Cambridge online dictionary)
     b. There's books on the desk. (Sobin, 1997, 320:(4))

In contrast, in French *il*-construction, the unaccusative verb never agrees with its postverbal argument, but rather shows agreement with the expletive (20). In SV constructions, lack of agreement is impossible (21).

(20) French

a. *Il est arrivé trois garçons.*
   it is arrived.M.SG three boy.M.PL
   'Three boys have arrived.'

b. **Il sont arrivés trois garçons.*
it are arrived.M.PL three boy.M.PL

(21) a. *Trois garçons sont arrivés.*
three boy.M.PL are arrived.M.PL
'Three boys arrived.'

b. **Trois garçons est arrivé.*
three boy.M.PL is arrived.M.SG

In Portuguese (22), unaccusative verbs appear in both SV and VS (Costa, 2001), whereas in Italian (23), unaccusative verbs typically appear in VS (Lambrecht, 1994, 17), with the verb normally agreeing with its subject. However, instances of lack of agreement are also attested, as in Hebrew, in VS only, as illustrated in Brazilian Portuguese (24) (Kato, 2002), European Portuguese (25) (Costa, 2001), and Tuscan Italian (26) (Nocentini, 1999).

(22) Portuguese

a. *Esses dois tipos apareceram.*
these.M two.M guy.M.PL appeared.3PL
'These two guys appeared.'

b. *Apareceram esses dois tipos.*
appeared.3PL these.M two.M guy.M.PL

(23) Italian

Sono arrivate le ragazze.
are arrived.F.PL the.F.PL girl.F.PL
'The girls have arrived.'

(24) Brazilian Portuguese

a. *Quando apareceu os 7 dragões.*
when appeared.3SG the.M.PL 7 dragon.M.PL
'When the 7 dragons appeared.' (`https://tinyurl.com/4dw3ctac`)

b. *Daí nasceu os rumores.*
from.there was.born.3SG the.M.PL rumor.M.PL
'The rumors originated from this.' (`https://tinyurl.com/w9t793yh`)

(25) European Portuguese

a. *Fechou muitas fábricas.*
closed.3SG many.F.PL factory.F.PL
'Many factories closed.' (Costa, 2001, 8:(19b))

b. *Chegou as cadeiras.*
arrived.3SG the.F.PL chair.F.PL
'The chairs arrived.' (Costa, 2001, 8:(19c))

(26) Tuscan Italian

a. *Con quest'umido nasce i funghi.*
with this.humid be.born.3SG the.M.PL mushroom.M.PL
'Mushrooms grow with this humidity.' (Nocentini, 1999, 319:(10))

b. *Stasera viene le tue amiche a trovar-ti?*
tonight comes.3SG the.F.PL your.F.PL friend.F.PL to find-you.SG
'Are your girlfriends coming to see you tonight?' (Nocentini, 1999, 319:(8))

In the above examples (24-26), which exhibit lack of agreement with the postverbal internal argument subject, nominative is not morphologically marked. This pattern lends some support to the proposed link between loss of agreement and the default nature of nominative. We now turn to the loss of postverbal pronominal subjects, and the loss of nominative in Hebrew VS constructions.

5.2 Pronominal subjects and accusative case

An independent general process of loss of postverbal pronominal subjects has occurred in Hebrew. In earlier stages of the language, specifically in Biblical Hebrew, pronominal subjects appeared postverbally as nominative subjects (in bold in 27).

(27) Biblical Hebrew

a. *wə-jɔsˁɔʔ me-ʕimm-ɔx* ***huʔ*** *wu-vɔnɔw ʕimm-o.*
and-exited.3M.SG from-within-you he and-sons.his with-him
'Then shall he go out from thee, he and his children with him.' (Leviticus 25:41; American Standard Version)

b. *wa-jjivraħ* ***huʔ*** *wə-xɔl ʔăšɛr lo.*
and-will.flee.3M.SG he and-all that to.him
'So he fled with all his possessions.' (Genesis 31:21; Berean Standard Bible)

Following the revival of Hebrew, which began at the end of the 19th century, these postverbal pronouns started disappearing. In contemporary Hebrew, pronouns appear preverbally, postverbal nominative pronouns are impossible (28). [12]

(28) a. **lo nišar hu.*
NEG remained.3M.SG he
'He/It did not remain.'

[12] The verb *nišar* is ambiguous between 'remain' and 'stay', the latter is ungrammatical in VS order, while the former is possible, as shown by the grammaticality of (12c) above and (30a) below, for instance. Postverbal pronominal subjects are only possible with focus, in particular, in the first and second person (e.g. *nafalta rak ATA* 'fell.2M.SG only YOU'). To our knowledge, no study has traced the origin of the process leading to the loss of nominative pronominal subjects.

b. **nigmar hu.*
finished.3M.SG he
'It is over.'

This disappearance process is not specific to unaccusatives. Recall that high-register Hebrew exhibits external postverbal subjects in the so-called triggered inversion construction (see section 2.1, example (8a)). Nominative pronouns have disappeared from such constructions as well (the more archaic the style of the text in which they appear is perceived by speakers, the better they sound) (29).

(29) *??/*ba-bkarim yašen hu.*
in.the-mornings sleep.M.SG he
'In the mornings he sleeps.'

This general loss of nominative posetverbal pronouns may have accelerated the loss of nominative case in the internal argument position of VS configurations.

Moreover, not only are postverbal nominative pronouns impossible, but these pronouns have even begun appearing in accusative case in the internal argument position of non-agreeing unaccusatives (30a-30b). This provides direct evidence for the loss of nominative in that position. In contrast, postverbal external subjects in triggered inversion constructions cannot be realized as accusative pronouns (31).

(30) a. *lo nišar oto.*
NEG remained.3M.SG him
'It did not remain.' (Conv.)

b. *ada'in lo nigmar li otam.*
still NEG finished.3M.SG DAT.me them
'They haven't run out yet.' (`https://tinyurl.com/4rrvv54w`)

(31) **ba-bkarim yašan oto.*
in.the-mornings slept.3M.SG him

Demonstrative nominative pronouns are also impossible (32a). They can only appear introduced by the accusative, direct object marker, *et* (32b), which typically precedes definite direct objects, including demonstratives and lexical noun phrases.

(32) a. **nišar le-xa ze.*
remained.3M.SG DAT-you this
Int.: 'You still have it.'

b. *nišar le-xa et ze be-mida 6?*
remained.3M.SG DAT-you ACC this in-size 6
'Do you still have it in size 6?' (`https://tinyurl.com/53t9m9hy`)

The direct object marker has also begun appearing with non-agreeing unaccusatives beyond the pronominal domain. Attested examples of unaccusatives with a postverbal lexical subject introduced by the accusative marker *et* are given in (33). [13] Halevy (2016) observes that the subject in such cases must be inanimate:

(33) a. *hofi'a li et ha-mila ha-zot ba-milon.*
appeared.3M.SG DAT.me ACC the-word.F.SG the-this.F.SG in.the-dictionary
'This word appeared in the dictionary.' (Conv.)

b. *nišar li et ha-sfaton-im ha-regil-im.*
remained.3M.SG DAT.me ACC the-lipstick-M.PL the-regular-M.PL
'I still have the regular lipsticks.' (Conv.)

c. *ala li et ha-maxšava ha-zot.*
arose.3M.SG DAT.me ACC the-thought.F.SG the-this.F.SG
'This thought came to my mind.' (Conv.)

d. *nišbar li et ha-lev.*
broke.3M.SG DAT.me ACC the-heart.M.SG
'My heart broke.' (`https://tinyurl.com/mr27j27j`)

e. *nisgar li et ha-ma'agal.*
closed.3M.SG DAT.me ACC the-circle.M.SG
'I got a closure.' (Conv.)

The occurrence of accusative postverbal subjects with unaccusatives supports our claim regarding the loss of nominative. Accusative is licensed only in the complement position – just like lack of ϕ-agreement – suggesting that both phenomena are possible only in the internal argument position. Since with other types of verbs in the triggered VS, the subject does not occupy the internal argument position, neither lack of ϕ-agreement nor accusative marking is possible.

The accusative marker *et* has already been argued to appear with (definite) noun phrases – both pronominal and lexical DPs (34) – in impersonal adjectival constructions where no other case is available (see Siloni (1997)).

[13] An anonymous reviewer notes that some instances of cooccurrence of agreement and accusative marking can be found:

(i) *nišaru/hayu li et kol ha-mesim-ot ha-ele.*
remained.3PL/were.3PL DAT.me ACC all the-task-F.PL the-these
'I have all these tasks left.'/'I had all these tasks.'

Indeed, we observed such cases, occurring predominantly with the quantifier *kol* 'all' introducing the postverbal argument. These instances may reflect hypercorrection, where speakers add accusative marking because they analyze the postverbal subject as an object, even though the verb agrees with it.

(34) a. *haya katuv šam et ze.*
was.3M.SG written.M.SG there ACC this
'This was written there.' (Conv.)

b. *lo haya katuv et ha-ša'a.*
NEG was.3M.SG written.M.SG ACC the-hour.F.SG
'The time was not written.' (Siloni, 1997)

c. *xaser li et ha-madrega ben-le-ven.*
missing.M.SG DAT.me ACC the-step.F.SG in.between
'I am missing the step in between.' (Conv.)

d. *lo haya mutkan šam et ha-font ha-ze.*
NEG was.3M.SG installed.M.SG there ACC the-font the-this
'This font was not installed there.' (Conv.)

5.3 From loss of nominative to loss of agreement

Nominative case is the prototypical example of unmarked case – morphologically and possibly structurally (see Neeleman and Weerman (2012)) as well as in case hierarchies à la Marantz (2000). Nominative has been closely associated with finiteness and morphological agreement, being a central exemplar of head–DP relationship, underlying case assignment (Vergnaud, 2008; Chomsky, 1981; George and Kornfilt, 1981) or Agree (Chomsky, 2000, 2001). It is the latter point that the present case study reinforces, as the loss of ϕ-agreement cooccurs with the loss of nominative, as evidenced by the emergence of accusative case (section 5.2). In what follows we argue that these phenomena cooccur because the loss of nominative has led to the loss of agreement.

Following many others, we assume that case is not necessary for nominal licensing (e.g. Bobaljik and Landau (2009); Marantz (2000); Preminger (2024)), but rather for rendering the noun phrase visible for Agree. Case features are uninterpretable (uCase). They enter the derivation unvalued. Their role is to activate the noun phrase (the goal), making it visible to a probe for Agree (Chomsky, 2000, 2001). Agree is thus a syntactic operation between two linguistic objects, the probe and the goal. (For further discussion of case, see section 5.5.)

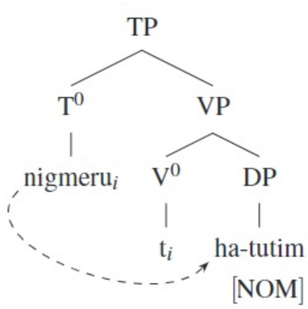
As discussed in section 1, agreement is optional in colloquial Hebrew in VS with unaccusatives.

(35) a. *nigmeru ha-tut-im.*
finished.3PL the-strawberry-M.PL
'There are no more strawberries.'

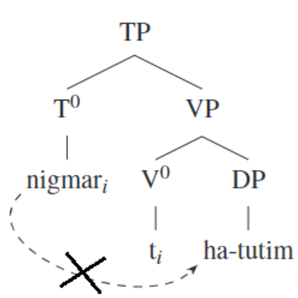
b. *nigmar* *ha-tut-im.*
finished.3M.SG the-strawberry-M.PL

We suggest that this optionality arises from speakers having access to two alternative analyses of VS configurations with unaccusatives: (i) the postverbal argument (the subject) bears a uCase feature, triggering Agree (36a); or (ii) the postverbal argument is caseless, that is, does not bear a uCase feature, and is therefore invisible to Agree, resulting in the absence of ϕ-agreement (36b). The convergence of two facts – that nominative subjects are morphologically unmarked and that the internal-argument position is typically non-nominative – enables speakers to analyze postverbal subjects as caseless in this position. Under analysis (i), Agree between T with unvalued uninterpretable ϕ-features, $T[_{u\phi}]$, and a uCase-bearing postverbal argument results in ϕ-agreement and nominative valuation, as illustrated in (36a) below. Under analysis (ii), the caseless postverbal argument is invisible to Agree and therefore not eligible to value $T[_{u\phi}]$, resulting in the absence of ϕ-agreement, as schematized in (36b).

(36) a. Structural representation of (35a)



b. Structural representation of (35b)



Arguably, the verb's default ϕ-feature set in (36b) reflects agreement with an externally merged null expletive *pro*, which also satisfies the EPP under both analyses (i-ii). Only a visible internal argument – that is, one bearing uCase – can raise to SpecTP; without case, the argument is invisible to the probe. Hence, lack of ϕ-agreement can only occur with postverbal subjects as raised subjects have been probed by $T[_{u\phi}]$. The stages leading to the lack of agreement in VS configurations with unaccusatives and the emergence of accusative case with some of them are summarized in (37):

(37) a. Loss of *u*Case on the (morphologically unmarked) subject
b. The postverbal argument becomes invisible to probing
c. Loss of ϕ–agreement with the argument due to its invisibility
d. Default ϕ–features surface, arguably, reflecting agreement with a null expletive
e. Emergence of accusative postverbal subjects with some unaccusatives

Further support for the relation between the loss of nominative case and the loss of agreement comes from European Portuguese, which exhibits agreementless VS configurations with unaccusatives (see (25) above). However, when the postverbal subject is a nominative pronoun rather than an unmarked lexical noun phrase, lack of agreement is impossible (38b) (Costa, 2001). In other words, lack of agreement can occur only when the subject can be analyzed as caseless.

(38) European Portuguese

a. *Chegaram eles.*
arrived.3PL they.NOM
'They arrived.'

b. **Chegou eles.*
arrived.3SG they.NOM (Costa, 2001, 12:(28))

Our findings show increased acceptability in structural intervention configurations, specifically in cases involving a possessive dative or an adverb, as compared to the other conditions (see Section 4.3). However, the acceptability ratings of the experimental conditions – even without the higher-rated possessive dative – do not differ significantly from those of the colloquial fillers, suggesting that they are all grammatical in low-registered varieties of Hebrew. Attested examples of verb-subject disagreement with no intervention point in the same direction. We thus conclude that intervention is not a prerequisite for grammaticality, but rather an extra-grammatical effect improving acceptability.

Psycholinguistic research has shown that intervening material can interfere with sentence processing, including agreement processing; that is, it can disrupt or increase the difficulty of establishing the relevant dependency. Various aspects of interference have been studied, including memory load, integration costs, and the similarity of the intervening material to the constituent(s) involved (e.g. Gibson (2000); Gordon et al. (2004); Lewis et al. (2006); Franck et al. (2006)). In light of this, intervening material between the verb and the postverbal subject may make agreement-less sentences more acceptable, since the relation between the verb and the subject may be less readily established than when the two are adjacent. The effect of similarity may underlie a potential preference for a possessive dative intervener, a nominal element that mirrors the categorial status of the postverbal subject.

In section 5.4, we argue that over the past century, Hebrew existential and possessive constructions, have undergone a parallel process to that summarized in (37).

5.4 Existential and possessive constructions

Hebrew existential and possessive constructions share the same structure, with the possessive construction adding a possessive dative, as illustrated in (39). In the present tense, the particles *yeš* and *eyn* are used to express existence or possession and non-existence or lack of possession, respectively. These particles are uninflected forms.

(39) a. *yeš/eyn* *mesiba* *be-šabat.*
exist/NEG.exist party.F.SG on-Saturday
'There is a/There is no party on Saturday.'

b. *yeš/eyn* *li* *mexonit.*
exist/NEG.exist DAT.me car.F.SG
'I (don't) have a car.'

In the past and future tenses, existential and possessive constructions use the verb 'be'. In earlier stages of Hebrew, agreement between the verb and its postverbal subject was obligatory.

(40) a. *hayta/tihiye* *mesiba* *be-šabat.*
was.3F.SG/will.be.3F.SG party.F.SG on-Saturday
'There was/will be a party on Saturday.'

b. *hayta/tihiye* *li* *mexonit.*
was.3F.SG/will.be.3F.SG DAT.me car.F.SG
'I had/will have a car.'

During the revival of Hebrew, these constructions began to exhibit lack of agreement (41), and the postverbal subject started to bear accusative case – phenomena documented as early as 1911 (Reshef, 2008). In Modern Hebrew, the postverbal subject can appear with accusative marking, whether as a pronoun (42a), a demonstrative (42b) or a lexical noun phrase (42c) (Berman, 1980; Melnik, 2006, 2018; Netz and Kuzar, 2011; Shlonsky, 1987; Taube, 2015; Ziv, 1976). Nominative postverbal pronominals, including demonstratives have become impossible (43a-43b), and a nominative definite lexical noun phrase now sounds archaic (43c). [14]

(41) a. *haya* *mesiba* *be-šabat.*
was.3M.SG party.F.SG on-Saturday
'There was a party on Saturday.'

b. *haya* *li* *mexonit.*
was.3M.SG DAT.me car.F.SG
'I had a car.'

(42) a. *haya* *(li)* *oto* *ba-sifriya.*
was.3M.SG (DAT.me) him in.the-library
'It was/(I had it) in the library.'

b. *haya* *(li)* *et* *ze* *ba-sifriya.*
was.3M.SG (DAT.me) ACC this in.the-library
'This was/(I had this) in the library.'

c. *haya* *(li)* *et* *ha-sefer* *ha-ze* *ba-sifriya.*
was.3M.SG (DAT.me) ACC the-book the-this in.the-library
'This book was in the library/(I had this book in the library).'

(43) a. **haya* *(li)* *hu* *ba-sifriya.*[15]
was.3M.SG (DAT.me) he in.the-library

b. **haya* *(li)* *ze* *ba-sifriya.*
was.3M.SG (DAT.me) this in.the-library

[14] Taube (2015) suggests that the object marking of definite noun phrases in existential and possessive constructions results from the influence of Yiddish, a contact language (the native language of many Eastern European Jews who revived Hebrew). In Yiddish negated existential constructions, a definite noun phrase can receive either accusative marking or nominative case (i).

(i) a. *ikh zukh khave-n,* *vu* *iz khave?*
I seek Khave-ACC where is Khave.NOM
'I seek Khave, where is Khave?'

-ništo *khave!*
NEG.exist Khave.NOM
'-No Khave.'

b. *ništo* *mikh.*
NEG.exist me.ACC
'I'm gone.'

In Russain, Polish, and also Ukranian (additional contact languages), the noun phrase is marked with genitive case. In Hebrew the genitive marker *šel* is limited to the nominal domain.

[15] As mentioned in note 12, postverbal nominative pronouns are possible with focus in the first and second person (e.g. *hayita (rak) ATA ba-xeder* 'was.2M.SG (only) YOU in.the-room'. When the verb does not show agreement, an accusative pronoun must be used (*haya (rak) otxa ba-xeder* 'was.3M.SG (only) you.ACC in.the-room').

c. *??haya ha-sefer ha-ze ba-sifriya.*
was.3M.SG the-book the-this in.the-library

Reasonably, this diachronic development represents an earlier process of the type that unaccusative VS constructions are currently undergoing in colloquial Hebrew. First, the postverbal subject is analyzable as caseless. Second, the loss of *u*Case makes the argument invisible to probing by T, and consequently, the constructions start exhibiting lack of agreement. Finally, the postverbal argument appears in accusative case. Today, lack of agreement in existential and possessive constructions has become dominant and the absence of the object marker is judged by native speakers as archaic, poetic, odd or impossible (43a-43c). Prescriptive attempts to resist this process have failed (Ben-Mordechai, 1943).

Thus, it appears that the process currently affecting unaccusative VS constructions mirrors the one existential and possessive constructions underwent in the previous century. In the case of existential and possessive constructions, both agreement and lack-of agreement analyses are possible with indefinite postverbal DPs. Definite postverbal DPs no longer trigger agreement, and are analyzed as direct objects, obligatorily introduced by the accusative marker *et*. In contrast, in unaccusative VS constructions, the insertion of the direct object marker *et* is (still) limited. This is expected as loss of ϕ–agreement with unaccusatives is recent, characterizing only the low registers of Hebrew, alongside the prevailing, standard agreeing construction. The emergence of accusative marking may (or may not) spread to additional unaccusatives in the future.

5.5 A note on Dependent Case Theory

As mentioned in section 5.3, our proposal relies on the view that *u*Case is required for the Agree operation, that is, for probing by the functional head T. A prominent alternative to the Agree approach is Marantz's (2000) Dependent Case Theory, which has been adopted by many scholars (Baker (2015); Bobaljik (2008); Preminger (2024), among others). This theory treats case as a postsyntactic phenomenon whose realization follows the disjunctive hierarchy below:

(44) Case realization disjunctive hierarchy (Marantz, 2000):

a. Lexically governed case

b. Dependent case

c. Unmarked case

d. Default case

The hierarchy suggests that lexical case – a case governed by a lexical head – is assigned first. In the absence of lexical case and presence of two unmarked noun phrases in the domain, one of them receives dependent case. In Nominative-Accusative systems, dependent accusative case is assigned to the lower noun phrase. In Ergative-Absolutive systems, the higher noun phrase, receives dependent ergative. If a noun phrase remains caseless, it receives unmarked case according to its domain, for instance, genitive in the DP domain, and nominative in the TP domain. Finally, if the DP is still caseless, default case is assigned as a last resort.

The cooccurrence of the loss of nominative and the loss of agreement in our data is straightforward under the Agree approach to nominative, but less so under Dependent Case Theory. On the latter view, the Hebrew nominative qualifies as both the unmarked case in the TP domain, and as the default case occurring on noun phrases in isolation (i.e. when no other case is available). As a default case, both its loss on the postverbal subject of unaccusatives is not expected as is its cooccurrence with the loss of agreement. Hebrew accusative is marked on (definite) direct objects, which qualifies it as dependent case. In addition, it appears in the complement position of some unaccusatives and impersonal adjectival constructions (neither set has a common denominator). Two options present themselves regarding the latter occurrences. One is that accusative in these constructions is a dependent case, which is assigned to the lower noun phrase in the presence of a null expletive acting as the higher noun phrase. However, if this were the case, we would expect accusative marking to appear systematically, rather than sporadically with certain adjectives and unaccusatives that fail to show agreement. Alternatively, one may suggest that accusative in Hebrew, in addition to being a dependent case, is a lexical case governed by some unaccusatives and adjectives. However, if these specific verbs and adjectives are endowed with lexical accusative, it is unclear why the realization of accusative coincides with the absence of agreement. Further exploring the implications of our data for Dependent Case Theory is beyond the scope of this paper.

6 Conclusion

The paper investigates an agreement asymmetry in colloquial Hebrew: unaccusative verbs optionally fail to exhibit ϕ-agreement with a postverbal subject, but not with a preverbal one. We conducted two experiments to examine this pattern more closely. The results indicate that in low-register varieties of Hebrew lack of agreement between an unaccusative verb and its postverbal subject is possible, and that intervention of a possessive dative or an adverb between the verb and its subject improves it. The findings clarify that this phenomenon of lack of agreement, which Preminger interpreted as evidence for his approach to grammaticality, does not in fact corroborate it.

We suggest that unaccusative constructions involving a postverbal subject are unstable with respect to case and, consequently, agreement. Specifically, in languages that do not morphologically mark nominative, such as Hebrew, a postverbal subject in the complement position may be analyzed by speakers as caseless. Two analyses can thus coexist: a case-bearing postverbal argument is visible to Agree, triggering ϕ-agreement, whereas a caseless postverbal argument is invisible to T, resulting in a lack of ϕ-agreement that is realized as default ϕ-features.

We offer some crosslinguistic evidence for the instability of agreement in such unaccusative VS configurations. Further, we present evidence for the loss of nominative – specifically, the emergence of accusative – and suggest that this loss in these agreementless constructions was aided by the broader independent loss of postverbal nominative pronouns. Finally, we show that the process currently undergone by unaccusative VS constructions mirrors the one that Hebrew existential and possessive constructions underwent in the previous century.

Abbreviations

ACC = accusative, CS = construct state, DAT = dative, F = feminine, INF = infinitive, M = masculine, NEG = negation, NOM = nominative, PASS = passive, PL = plural, SG = singular

Data availability

All supplementary files including the appendix, materials and collected data are available on OSF at `https://osf.io/f6q45/?view_only=1419e575e8a942ccbeaaf1bec41c82a6`.

Ethics and consent

The research received the approval of the University Institutional Review Board Ethics Committee (proposal no. 0004580-1). All participants in our experiments gave their informed consent before participating.

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Competing interests

The authors have no competing interests to declare.

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